

## Paper 2 - Set 01 - Solutions

Generated practice paper: answer key and worked solutions.

### Answer key

| Question | Answer | Topic                                      |
|----------|--------|--------------------------------------------|
| 1        | B      | Differentiation                            |
| 2        | C      | Triangles, radians and sectors             |
| 3        | F      | Polynomials, factor and remainder theorems |
| 4        | C      | The logic of arguments                     |
| 5        | E      | Pythagoras, mensuration and 3D             |
| 6        | D      | The logic of arguments                     |
| 7        | A      | Mathematical proof                         |
| 8        | F      | Statistics                                 |
| 9        | E      | Mathematical proof                         |
| 10       | A      | Identifying errors in proofs               |
| 11       | F      | Probability                                |
| 12       | F      | Functions and transformations              |
| 13       | A      | Identifying errors in proofs               |
| 14       | A      | Identifying errors in proofs               |
| 15       | H      | Binomial expansion                         |
| 16       | G      | The logic of arguments                     |
| 17       | E      | Trigonometric functions and equations      |
| 18       | E      | Exponentials and logarithms                |
| 19       | A      | Mathematical proof                         |
| 20       | F      | Chains of reasoning                        |

## Worked solutions

### Question 1 DIF-06 · Differentiation · spec MM6.1

The displacement  $s$  metres of a particle at time  $t$  seconds is  $s = t^3 - 6t^2 + 9t$ . At what times is the particle momentarily at rest?

**Answer: B**

$$\text{Velocity} = \frac{ds}{dt} = 3t^2 - 12t + 9 = 3(t - 1)(t - 3) = 0 \text{ at } t = 1 \text{ and } t = 3.$$

#### Full method - how and why

'At rest' means velocity zero, and velocity is the derivative of displacement.

1 Differentiate the displacement:  $v = \frac{ds}{dt} = 3t^2 - 12t + 9$ .

why: Velocity is the rate of change of displacement.

2 Factorise:  $3(t^2 - 4t + 3) = 3(t - 1)(t - 3)$ .

why: Take the 3 out first.

3 Set  $v = 0$ :  $t = 1$  and  $t = 3$ .

why: 'Momentarily at rest' is exactly  $v = 0$ .

#### Where it goes wrong

- Setting  $s = 0$  instead of  $v = 0$ , which gives  $t = 0$  and  $t = 3$  - an offered option.
- Forgetting one of the two roots.

**Takeaway** Displacement  $\rightarrow$  velocity  $\rightarrow$  acceleration by differentiating; 'at rest' means the velocity is zero, not the displacement.

### Question 2 TRI-03 · Triangles, radians and sectors · spec MM4.2

A sector of a circle of radius 6 cm has area  $15\pi \text{ cm}^2$ . Find the arc length of the sector.

**Answer: C**

$$\frac{1}{2}r^2\theta = 18\theta = 15\pi \Rightarrow \theta = \frac{5\pi}{6}. \text{ Arc} = r\theta = 6 \cdot \frac{5\pi}{6} = 5\pi.$$

#### Full method - how and why

Sector area gives the angle; the angle gives the arc length. Both formulas need radians.

1 Use the sector area formula  $A = \frac{1}{2}r^2\theta$ :  $\frac{1}{2}(36)\theta = 18\theta$ .

why:  $r = 6$ , so  $\frac{1}{2}r^2 = 18$ . The formula is only valid with  $\theta$  in radians.

2 Set equal to the given area:  $18\theta = 15\pi$ , so  $\theta = \frac{5\pi}{6}$ .

why: Divide:  $\frac{15\pi}{18} = \frac{5\pi}{6}$ .

3 Use the arc-length formula  $s = r\theta = 6 \times \frac{5\pi}{6} = 5\pi \text{ cm}$ .

why: The 6 cancels neatly - a sign the angle is right.

#### Where it goes wrong

- Quoting the angle  $\frac{5\pi}{6}$  as the arc length - it is offered as a distractor.
- Using  $A = r^2\theta$  without the  $\frac{1}{2}$ , which halves the angle and gives  $\frac{5\pi}{2}$ .

**Takeaway**  $A = \frac{1}{2}r^2\theta$  and  $s = r\theta$  (radians); find  $\theta$  from one, then substitute into the other.

### Question 3 POL-01 · Polynomials, factor and remainder theorems · spec MM1.6

When  $x^3 - 4x^2 + kx + 6$  is divided by  $(x - 3)$  the remainder is  $-6$ . Find  $k$ .

**Answer: F**

Remainder Theorem:  $27 - 36 + 3k + 6 = -6 \Rightarrow 3k - 3 = -6 \Rightarrow k = -1$ .

#### Full method - how and why

A remainder is given, so use the Remainder Theorem: substitute the root of the divisor and set the result equal to the remainder.

- 1 The divisor  $(x - 3)$  is zero at  $x = 3$ .

why: The Remainder Theorem says the remainder on dividing by  $(x - a)$  is  $f(a)$ ; you need the value of  $x$  that makes the divisor vanish.

- 2 Substitute:  $f(3) = 27 - 36 + 3k + 6 = 3k - 3$ .

why: Work the numerical terms out first ( $27 - 36 + 6 = -3$ ) so only the  $k$  term is left to carry.

- 3 Set equal to the given remainder:  $3k - 3 = -6$ .

why: This is what the Remainder Theorem gives you - an equation, not an identity.

- 4 Solve:  $3k = -3$ , so  $k = -1$ .

why: Simple linear equation.

#### Where it goes wrong

- Substituting  $x = -3$  (the sign of the root of  $x - 3$  is  $+3$ ), which gives a different value of  $k$ .
- Setting  $f(3) = 0$  out of habit - that is the Factor Theorem, and only applies when the remainder is zero.

**Takeaway** Divisor  $(x - a) \Rightarrow$  substitute  $x = a$ ; the result is the remainder, which is zero only when  $(x - a)$  is a factor.

### Question 4 LGC-07 · The logic of arguments · spec Arg2

Let  $P$ : 'the quadrilateral  $ABCD$  is a square' and  $Q$ : 'the quadrilateral  $ABCD$  has four equal sides'. Which one of the following is true?

**Answer: C**

A square has four equal sides ( $P \Rightarrow Q$ , sufficient), but a rhombus that is not a square shows  $Q$  does not imply  $P$  (not necessary).

#### Full method - how and why

The same two-direction test, now in geometry. Look for a shape satisfying one condition but not the other.

- 1 Sufficiency: every square has four equal sides, so  $P \Rightarrow Q$ . **True.**

why: Straight from the definition of a square.

- 2 Necessity: does four equal sides force a square? A rhombus with angles  $60^\circ$  and  $120^\circ$  has four equal sides and is not a square.

why: A square also needs right angles; equal sides alone do not fix the angles.

- 3 So  $P$  is sufficient but not necessary for  $Q$ .

why:  $P$  is the stronger condition, so it implies the weaker one but not conversely.

#### Where it goes wrong

- Confusing which is the stronger condition: the more demanding property (square) is sufficient for the less demanding one (four equal sides).
- Failing to produce a concrete counterexample and guessing.

**Takeaway** A stronger condition is sufficient for a weaker one; the weaker is necessary for the stronger. Name a counterexample to confirm.

**Question 5** MEN-11 · Pythagoras, mensuration and 3D · spec M5.16

Which of the following statements is/are true?

- I Doubling the radius of a circle doubles its circumference.
- II Doubling the radius of a sphere multiplies its volume by 8.
- III Doubling the radius of a cylinder while halving its height leaves its volume unchanged.

**Answer: E**

$C = 2\pi r$  is linear in  $r$  (I).  $V \propto r^3$  (II).  $\pi(2r)^2 \cdot \frac{h}{2} = 2\pi r^2 h$ : doubled, not unchanged (III false).

**Full method - how and why**

Three scaling claims. Each is checked by substituting the new dimensions into the formula.

- 1 I:  $C = 2\pi r$  is linear in  $r$ , so doubling  $r$  doubles  $C$ . **True.**  
why: Circumference scales with the first power of the radius.
- 2 II:  $V = \frac{4}{3}\pi r^3$ , so doubling  $r$  multiplies the volume by  $2^3 = 8$ . **True.**  
why: Volume scales with the cube.
- 3 III: new volume  $= \pi(2r)^2 \left(\frac{h}{2}\right) = \pi(4r^2) \left(\frac{h}{2}\right) = 2\pi r^2 h$  - **double**, not unchanged. **False.**  
why: The radius is squared, so doubling it quadruples that factor; halving the height only removes a factor of 2.
- 4 I and II only.  
why: Match to the option list.

**Where it goes wrong**

- Assuming the doubling and halving in III cancel - the radius is squared, so they do not.
- Confusing area and volume scaling in II.

**Takeaway** Substitute the new dimensions into the formula; squared quantities change faster than linear ones.

**Question 6** LGC-21 · The logic of arguments · spec Arg3

Consider the statement  $S$ : \*\*for all real  $x$ ,  $x^2 - 6x + 10 > 0$ .\*\* Which of the following is true?

**Answer: D**

Completing the square shows the expression is at least 1 for every  $x$ .

**Full method - how and why**

A universal claim about a quadratic. Completing the square settles it, and the question also tests whether the **reason** given is sound.

- 1 Complete the square:  $x^2 - 6x + 10 = (x - 3)^2 - 9 + 10 = (x - 3)^2 + 1$ .  
why: Halve the coefficient of  $x$  to get  $-3$ , then correct by  $-(-3)^2 = -9$ .
- 2  $(x - 3)^2 \geq 0$ , so the expression is  $\geq 1 > 0$  for every real  $x$ . **True.**  
why: A square is never negative; the minimum value is 1 at  $x = 3$ .
- 3 Check the false reasons:  $x = 3$  gives  $9 - 18 + 10 = 1$ , not 0; and the discriminant  $36 - 40 = -4 < 0$ , so there are **no** real roots.  
why: The claim that it has two real roots is simply wrong, and would in any case not prove positivity.

**Where it goes wrong**

- Assuming any quadratic takes negative values - one with a positive leading coefficient and negative discriminant never does.
- Mis-completing the square as  $(x - 3)^2 + 10$  or  $(x - 6)^2 + 1$ .

**Takeaway** Completing the square proves positivity outright:  $a(x - h)^2 + k$  with  $a > 0, k > 0$  is always positive.

### Question 7 X2-27 · Mathematical proof · spec Prf1

Consider the claim: **\*\*if  $n$  is a positive integer then  $n^2 + n + 1$  is odd.\*\*** Which of the following is a valid proof?

**Answer: A**

A is a direct general argument. B only checks cases; C is irrelevant; D is correct for odd  $n$  but says nothing about even  $n$ ; E does not reach a contradiction.

#### Full method - how and why

Same 'which is a valid proof' pattern as PRF-01: general argument versus cases, irrelevance and incompleteness.

- 1 The valid proof:  $n^2 + n = n(n + 1)$  is a product of consecutive integers, hence even; adding 1 gives an odd number.

why: General, covering every positive integer with no case split.

- 2 Checking  $n = 1, 2, 3$  gives 3, 7, 13 - evidence, not proof.

why: Three cases out of infinitely many.

- 3 'It cannot be factorised, so it is odd' - factorisability has nothing to do with parity ( $6 = 2 \times 3$  and  $9 = 3 \times 3$  are respectively even and odd).

why: An irrelevant premise cannot support the conclusion.

- 4 'If  $n$  is odd then ... so  $n^2 + n + 1$  is odd' - correct but only for odd  $n$ ; even  $n$  is untreated. Incomplete.

why: A case-based proof must cover every case.

- 5 'Suppose  $n^2 + n + 1$  is even. Then  $n$  is even, contradiction' -  $n$  being even is not a contradiction;  $n$  can be even.

why: A contradiction proof must actually reach an impossibility.

#### Where it goes wrong

- Accepting the odd- $n$  argument because every line in it is true - truth is not completeness.
- Accepting the contradiction attempt because it uses the right vocabulary.

**Takeaway**  $n(n + 1)$  is always even, so  $n^2 + n + 1$  is always odd. A proof must be general, relevant and complete.

### Question 8 STA-05 · Statistics · spec M6.4

Which of the following statements about a scatter graph showing strong positive correlation between hours of revision and exam mark is/are valid?

- I Students who revised more tended to score higher.
- II Revising more causes a higher mark.
- III A line of best fit could be used to estimate the mark of a student who revised for a number of hours within the range of the data.

**Answer: F**

Correlation describes a tendency (I) and supports interpolation within the data range (III), but does not by itself establish causation (II).

#### Full method - how and why

A correlation-and-causation statement test. Decide what correlation does and does not licence.

- 1 I: strong positive correlation means higher revision went with higher marks. **True** - that is exactly what correlation describes.

why: 'Tended to' is a statement about the trend, not about every student.

- 2 II: correlation does not establish causation. **False.**

why: A hidden variable (motivation, prior ability) could drive both, so the data alone cannot prove cause.

- 3 III: using a line of best fit **within** the data range is interpolation. **True.**

why: Interpolation is legitimate; extrapolation beyond the data is what is unreliable.

- 4 So I and III only.

why: Match to the list.

#### Where it goes wrong

- Accepting II because the causal story sounds plausible - plausibility is not evidence.
- Rejecting III by confusing interpolation with extrapolation.

**Takeaway** Correlation licenses description and interpolation; it never licenses a causal claim on its own.

### Question 9 PRF-04 · Mathematical proof · spec Prf1

A student wants to prove that \*\*the square of any odd integer leaves remainder 1 when divided by 8\*\*. Which of the following is the most appropriate first step?

**Answer: E**

A direct proof:  $(2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$ , and  $k(k + 1)$  is even, so the square is  $8m + 1$ . Checking cases is not a proof; assuming the claim is circular.

#### Full method - how and why

Choosing a proof strategy. For a 'for all odd integers' claim, the standard opening is an algebraic parametrisation.

**1** Represent an arbitrary odd integer as  $2k + 1$ ,  $k$  an integer.

*why:* This one expression covers **every** odd integer, which is what a universal claim needs.

**2** (For context, the proof then runs:  $(2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$ .)

*why:* Factorising  $4k^2 + 4k$  as  $4k(k + 1)$  exposes the consecutive pair.

**3**  $k(k + 1)$  is a product of consecutive integers, hence even, say  $2m$ , so the square is  $8m + 1$ .

*why:* That is remainder 1 on division by 8, as claimed.

#### Where it goes wrong

- Checking 1, 3, 5, 7, 9 - evidence, not proof.
- Writing the odd integer as  $2k$ , which is the form of an **even** integer.
- Assuming the claim is true - circular.
- Reaching for contradiction ('assume the remainder is not 1'): legal, but here a direct proof is simpler, and the question asks for the most appropriate step.

**Takeaway** Universal claims about odd/even numbers start by writing the general form:  $2k$  or  $2k + 1$ .

**Question 10** ERR-09 · Identifying errors in proofs · spec Err2

A student proves that the equation  $x^2 + 2x + 3 = 0$  has no real solutions.

- I Complete the square:  $x^2 + 2x + 3 = (x + 1)^2 + 2$ .
- II  $(x + 1)^2 \geq 0$  for all real  $x$ .
- III So  $(x + 1)^2 + 2 \geq 2 > 0$  for all real  $x$ .
- IV Hence the equation has no real solutions.

Which of the following is correct?

**Answer: A**

Every step is valid.

**Full method - how and why**

Another valid proof. Completing the square and bounding below is a standard, correct technique.

- 1 Line I:  $(x + 1)^2 = x^2 + 2x + 1$ , so  $x^2 + 2x + 3 = (x + 1)^2 + 2$  ✓.  
why: Half the coefficient of  $x$  is 1; correct by  $3 - 1 = 2$ .
- 2 Line II: any real square is  $\geq 0$  ✓.  
why: True for every real  $x$ .
- 3 Line III: adding 2 gives  $\geq 2$ , and  $2 > 0$  ✓.  
why: Adding a constant to both sides of an inequality is always valid.
- 4 Line IV: an expression that is always at least 2 can never equal 0, so there are no real roots ✓.  
why: That is the definition of having no real solutions. (Check: discriminant  $4 - 12 = -8 < 0$  ✓.)

**Where it goes wrong**

- Claiming  $(x + 1)^2$  can be negative - not for real  $x$ .
- Claiming 'a positive expression can still equal zero' - it cannot; that is what positive means.
- Assuming there must be an error.

**Takeaway** Completing the square gives a rigorous lower bound; if the minimum is positive, there are no real roots.

**Question 11** PRO-03 · Probability · spec M7.7

$P(A) = 0.6$ ,  $P(B) = 0.5$  and  $P(A \text{ and } B) = 0.3$ . Which of the following statements is/are true?

- I A and B are independent.
- II A and B are mutually exclusive.
- III  $P(A \text{ or } B) = 0.8$

**Answer: F**

$P(A)P(B) = 0.3 = P(A \text{ and } B)$ , so independent (I). Not mutually exclusive since they can both happen (II false).  
 $P(A \text{ or } B) = 0.6 + 0.5 - 0.3 = 0.8$  (III).

**Full method - how and why**

Three definitions to test: independence, mutual exclusivity and the addition rule.

- 1 I: independence means  $P(A \cap B) = P(A)P(B)$ . Here  $0.6 \times 0.5 = 0.3$ , which matches. **True.**  
why: Independence is a numerical condition, not something you can eyeball.
- 2 II: mutually exclusive means  $P(A \cap B) = 0$ . It is  $0.3 \neq 0$ . **False.**  
why: They overlap, so they can happen together.
- 3 III:  $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.3 = 0.8$ . **True.**  
why: Subtract the overlap or you count it twice.
- 4 So I and III only.  
why: Match to the list.

**Where it goes wrong**

- Treating independent and mutually exclusive as the same thing - they are near-opposites: two events with non-zero probabilities cannot be both.
- Forgetting the  $-P(A \cap B)$  and answering 1.1, which is impossible for a probability.

**Takeaway** Independent:  $P(A \cap B) = P(A)P(B)$ . Mutually exclusive:  $P(A \cap B) = 0$ . Check the numbers, not the words.

**Question 12** FUN-12 · Functions and transformations · spec MM1.7

Let  $f(x) = \sqrt{x-3}$  for  $x \geq 3$  and  $g(x) = x^2 + 3$ . Which of the following statements is/are true?

- I  $g(f(x)) = x$  for all  $x \geq 3$
- II  $f(g(x)) = x$  for all real  $x$
- III  $f(g(x)) = |x|$  for all real  $x$

**Answer: F**

I:  $g(f(x)) = (\sqrt{x-3})^2 + 3 = x$  for  $x \geq 3$ : true.  $f(g(x)) = \sqrt{x^2 + 3} = |x|$ , so II fails (e.g.  $x = -1$ ) and III is true.

**Full method - how and why**

Two functions that almost invert each other. The asymmetry comes from  $\sqrt{x^2} = |x|$  and from the restricted domain.

**1** I:  $g(f(x)) = (\sqrt{x-3})^2 + 3 = (x-3) + 3 = x$  for  $x \geq 3$ . **True.**

*why:* Squaring a square root recovers the inside, provided the inside is non-negative - guaranteed by the stated domain  $x \geq 3$ .

**2** Compute the other order:  $f(g(x)) = \sqrt{(x^2 + 3) - 3} = \sqrt{x^2} = |x|$ .

*why:* Do the algebra once and read both remaining statements off it.

**3** II claims this equals  $x$  for all real  $x$ : false at  $x = -1$ , where  $|x| = 1 \neq -1$ . **False.**

*why:* The square root returns the non-negative value, so it cannot reproduce a negative input.

**4** III says it equals  $|x|$ , which is exactly what we found. **True.**

*why:* Same computation, correct conclusion.

**5** I and III only.

*why:* Match to the options.

**Where it goes wrong**

- Assuming  $f(g(x)) = x$  because  $f$  and  $g$  'look like' inverses - inverses require matching domains and ranges.
- Writing  $\sqrt{x^2} = x$ .

**Takeaway**  $f(g(x)) = x$  and  $g(f(x)) = x$  are different statements;  $\sqrt{x^2} = |x|$  is what breaks the symmetry here.

**Question 13** ERR-11 · Identifying errors in proofs · spec Err1

Claim:  $\int_{-1}^1 \frac{1}{x^2} dx = -2$ .

- I An antiderivative of  $x^{-2}$  is  $-x^{-1}$ .  
II So the integral equals  $\left[-\frac{1}{x}\right]_{-1}^1 = -1 - 1 = -2$ .  
III Hence the value is  $-2$ .

Which of the following is correct?

**Answer: A**

The integrand is positive everywhere it is defined, so a negative answer is a warning sign; the function is not defined on the whole interval and the integral does not exist in the ordinary sense.

**Full method - how and why**

The fundamental theorem of calculus needs the integrand to be defined (and continuous) throughout the interval.

- 1 Spot the warning sign:  $\frac{1}{x^2} > 0$  wherever it is defined, yet the 'answer' is negative.  
*why:* An integral of a positive function over an interval cannot be negative - the sign alone signals an error.
- 2 Line I is fine:  $\frac{d}{dx}(-x^{-1}) = x^{-2}$  ✓.  
*why:* Differentiate:  $-(-1)x^{-2} = x^{-2}$ .
- 3 Line II applies the fundamental theorem across  $[-1, 1]$ . But  $\frac{1}{x^2}$  is undefined at  $x = 0$ , which lies **inside** the interval. **First error.**  
*why:* The antiderivative is not continuous on the whole interval, so the theorem does not apply.
- 4 The arithmetic in line II is itself right ( $-\frac{1}{1} - (-\frac{1}{-1}) = -1 - 1 = -2$ ); the fault is applying the theorem at all.  
*why:* Correct arithmetic on an invalid application still gives nonsense.

**Where it goes wrong**

- Trying to 'fix' the sign to  $+2$  - the integral does not exist in the ordinary sense, so no finite value is right.
- Blaming line I; the antiderivative is correct.
- Ignoring the discontinuity because the limits themselves are perfectly ordinary numbers.

**Takeaway** Before applying  $\int_a^b f = F(b) - F(a)$ , check  $f$  is defined on all of  $[a, b]$ ; and a positive integrand can never give a negative integral.

**Question 14** ERR-07 · Identifying errors in proofs · spec Err2

A student writes the following argument.

**I**  $0.1 > 0.01$ .

**II** Taking logarithms to base 10 of both sides:  $-1 > -2$ .

**III** Multiplying both sides by  $\log_{10} \frac{1}{2}$ , which is negative, reverses the inequality:  
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$ .

Which of the following is correct?

**Answer: A**

Logarithms are increasing functions, so line II is valid;  $\log_{10} \frac{1}{2} < 0$ , so reversing the inequality in line III is correct.

### Full method - how and why

Sometimes the argument is simply valid. Check each step against the rules rather than assuming a trap.

**1** Line I:  $0.1 > 0.01$  ✓.

why:  $\frac{1}{10} > \frac{1}{100}$ .

**2** Line II:  $\log_{10}$  is an **increasing** function, so it preserves the inequality:  $\log_{10} 0.1 = -1$  and  $\log_{10} 0.01 = -2$ , and indeed  $-1 > -2$  ✓.

why: Applying an increasing function to both sides keeps the direction;  $\log_{10} 10^{-1} = -1$ .

**3** Line III:  $\log_{10} \frac{1}{2} < 0$  because  $\frac{1}{2} < 1$ . Multiplying an inequality by a negative number reverses it, which is exactly what is written ✓.

why:  $-1 > -2$  becomes  $-1 \times c < -2 \times c$  for  $c < 0$ .

**4** So all three lines are correct.

why: The question is testing whether you can certify a valid argument, not only spot faults.

### Where it goes wrong

- Assuming logarithms reverse inequalities - they do not;  $\log$  is increasing on its domain.
- Thinking  $\log_{10} \frac{1}{2}$  is positive; logs of numbers below 1 are negative.
- Misreading  $\log_{10} 0.1$  as 1 rather than  $-1$ .
- Hunting for an error simply because the question implies one might exist.

**Takeaway** Increasing functions preserve inequalities; multiplying by a negative reverses them.  $\log_b x < 0$  exactly when  $0 < x < 1$  (for  $b > 1$ ).

**Question 15** BIN-09 · Binomial expansion · spec MM2.4

Which of the following statements is/are true?

- I**  $\binom{n}{r} = \binom{n}{n-r}$  for all integers  $0 \leq r \leq n$   
**II**  $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$  for all integers  $0 \leq r < n$   
**III**  $\sum_{r=0}^n \binom{n}{r} = 2^n$  for all positive integers  $n$

**Answer: H**

I is the symmetry of Pascal's triangle; II is Pascal's rule (each entry is the sum of the two above it); III is  $(1+1)^n$  expanded. All true.

**Full method - how and why**

Three standard properties of binomial coefficients. Each has a one-line reason and a one-line check.

- I:**  $\binom{n}{r} = \binom{n}{n-r}$  - choosing  $r$  objects is the same as choosing the  $n-r$  to leave. **True.**  
*why:* Visible as the left-right symmetry of each row of Pascal's triangle.
- II:**  $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$  - Pascal's rule, i.e. each entry is the sum of the two above it. **True.**  
*why:* Check with numbers:  $\binom{3}{2} + \binom{3}{3} = 3 + 1 = 4 = \binom{4}{3}$  ✓.
- III:**  $\sum_{r=0}^n \binom{n}{r} = 2^n$  - put  $x = 1$  in  $(1+x)^n$ . **True.**  
*why:* Equivalently, a set with  $n$  elements has  $2^n$  subsets, counted by size.
- All three: 'I, II and III'.**  
*why:* Match to the options.

**Where it goes wrong**

- Doubting Pascal's rule because of the index shift on the right - test it with small numbers.
- Thinking the sum in III is  $2^{n+1}$  by miscounting the  $n+1$  terms.

**Takeaway** Symmetry, Pascal's rule and row sum  $2^n$  are the three facts about  $\binom{n}{r}$  worth knowing cold.

**Question 16** LGC-16 · The logic of arguments · spec Arg3

Which of the following statements is/are true about real numbers?

- I** For all  $x$  and all  $y$ , if  $x < y$  then  $x^2 < y^2$ .  
**II** For all  $x$  there exists  $y$  such that  $x < y$  and  $x^2 < y^2$ .  
**III** There exist  $x$  and  $y$  with  $x < y$  and  $x^2 > y^2$ .

**Answer: G**

I fails for  $x = -2$ ,  $y = 1$ . II: take  $y = |x| + 1$ . III:  $x = -2$ ,  $y = 1$  works.

**Full method - how and why**

Three quantified statements about squaring and ordering. Negative numbers decide all three.

- I:**  $x < y \Rightarrow x^2 < y^2$  for all reals? Take  $x = -2$ ,  $y = 1$ :  $-2 < 1$  but  $4 > 1$ . **False.**  
*why:* Squaring is only order-preserving for non-negative numbers.
- II:** for each  $x$ , is there  $y$  with  $x < y$  and  $x^2 < y^2$ ? Take  $y = |x| + 1$ : then  $y > x$  and  $y > |x|$  so  $y^2 > x^2$ . **True.**  
*why:* Choosing  $y$  larger than  $|x|$  guarantees both conditions at once - and  $y$  may depend on  $x$ .
- III:** do some  $x < y$  have  $x^2 > y^2$ ? The counterexample from I works:  $x = -2$ ,  $y = 1$ . **True.**  
*why:* Reuse the work - the same pair settles both.
- So II and III.**  
*why:* Match to the list.

**Where it goes wrong**

- Accepting I by testing only positive numbers.
- Rejecting II by picking a bad  $y$ ; you get to choose  $y$ , so choose a helpful one.

**Takeaway**  $x < y \Rightarrow x^2 < y^2$  needs  $0 \leq x$ ; and to prove  $\forall x \exists y$ , construct  $y$  as a formula in  $x$ .

### Question 17 TRG-05 · Trigonometric functions and equations · spec MM4.4

Which of the following statements about the graph of  $y = \cos x$  is/are true?

- I It is symmetric about the y-axis.
- II It is a translation of the graph of  $y = \sin x$  by  $\frac{\pi}{2}$  in the negative  $x$ -direction.
- III It has period  $\pi$ .

**Answer: E**

$\cos(-x) = \cos x$  so I is true.  $\sin(x + \frac{\pi}{2}) = \cos x$ , so II is true. The period of  $\cos$  is  $2\pi$ , so III is false.

#### Full method - how and why

Three properties of the cosine graph. Each is a standard fact; the false one is about the period.

- 1 I:  $\cos(-x) = \cos x$ , so the graph is unchanged by reflection in the y-axis. **True.**  
why: Cosine is an even function - its graph is symmetric about the y-axis, unlike sine.
- 2 II:  $\sin(x + \frac{\pi}{2}) = \cos x$ , and replacing  $x$  by  $x + \frac{\pi}{2}$  shifts the graph  $\frac{\pi}{2}$  to the left. **True.**  
why: Check a point:  $\sin$  at  $x = \frac{\pi}{2}$  equals 1, and  $\cos$  reaches 1 at  $x = 0$  - shifted left by  $\frac{\pi}{2}$  ✓.
- 3 III: the period of  $\cos$  is  $2\pi$ , not  $\pi$ . **False.**  
why:  $\cos(x + \pi) = -\cos x$ , so a shift of  $\pi$  flips the graph rather than repeating it.
- 4 I and II only.  
why: Match to the option list.

#### Where it goes wrong

- Thinking the period is  $\pi$  because  $\cos$  returns to zero every  $\pi$  - the **whole** shape must repeat.
- Getting the direction of the shift in II backwards.

**Takeaway**  $\cos$  is even with period  $2\pi$ ;  $\cos x = \sin(x + \frac{\pi}{2})$  is a shift to the left.

### Question 18 X2-06 · Exponentials and logarithms · spec MM5.2

Which of the following is a counterexample to the claim **\*\*'**for all positive  $x$  and  $y$ ,  $\log_{10}(x + y) \leq \log_{10} x + \log_{10} y$ **\*\*\***?

**Answer: E**

$x = y = 1$ :  $\log 2 > 0 = \log 1 + \log 1$ , so the inequality fails. For the others  $x + y \leq xy$  and the claim holds (log is increasing).

#### Full method - how and why

To break an inequality claim, look for the case where the two sides are closest - here small values, where logs go to zero or negative.

- 1 Test  $x = y = 1$ : left side  $\log_{10} 2 \approx 0.30$ ; right side  $\log_{10} 1 + \log_{10} 1 = 0$ .  
why:  $\log 1 = 0$  for any base, so the right side collapses.
- 2  $0.30 > 0$ , so the claimed inequality  $\leq$  fails. **Counterexample.**  
why: Hypothesis satisfied ( $x, y$  positive), conclusion false.
- 3 Why the others work:  $\log x + \log y = \log(xy)$ , and log is increasing, so the claim is just  $x + y \leq xy$ . For  $x = y = 2$ :  $4 \leq 4$  ✓;  $x = y = 4$ :  $8 \leq 16$  ✓;  $x = y = 5$ :  $10 \leq 25$  ✓;  $x = 10, y = 100$ :  $110 \leq 1000$  ✓.  
why: Converting the log inequality into an algebraic one makes all five cases quick to check.

#### Where it goes wrong

- Trying large values, where  $xy$  dwarfs  $x + y$  and the claim comfortably holds.
- Forgetting  $\log_{10} 1 = 0$  and miscomputing the right side.
- Thinking  $\log(x + y) = \log x + \log y$  - that is the error the whole question is built on.

**Takeaway**  $\log x + \log y = \log(xy)$ , never  $\log(x + y)$ . Test inequality claims near the boundary, where the sides are closest.

### Question 19 PRF-10 · Mathematical proof · spec Prf1

Consider the claim: **\*\*if  $x$  and  $y$  are real numbers with  $x + y > 10$ , then  $x > 5$  or  $y > 5$ .\*\*** Which of the following proves it?

**Answer: A**

A is proof by contrapositive/contradiction. B assumes the conclusion, C is an example, D proves something false, E's conclusion ('both exceed 5') is wrong.

#### Full method - how and why

Proving 'if  $P$  then  $(A \text{ or } B)$ '. Assuming both  $A$  and  $B$  fail and deriving a contradiction is the natural route.

**1** Negate the conclusion: 'not  $(x > 5 \text{ or } y > 5)$ ' means  $x \leq 5$  **and**  $y \leq 5$  (De Morgan).

*why:* Negating an 'or' gives an 'and' - and it gives you two usable facts instead of a disjunction.

**2** Add the inequalities:  $x + y \leq 10$ .

*why:* Inequalities of the same direction may be added.

**3** That contradicts the hypothesis  $x + y > 10$ , so the conclusion must hold.

*why:* Hypothesis true and conclusion false is impossible, which is exactly what proves the implication.

#### Where it goes wrong

- 'If  $x > 5$  then the conclusion holds' - this assumes part of what is to be proved.
- Testing  $x = y = 6$  - a single example proves nothing about all  $x, y$ .
- 'Suppose  $x > 5$ , then  $y > 10 - x$ , so  $y > 5$ ' - the algebra is wrong and it proves a false statement (both exceeding 5).
- 'The average exceeds 5 so both exceed 5' - the average exceeding 5 only forces **at least one** to.

**Takeaway** To prove an 'or', negate it into an 'and' and look for a contradiction - De Morgan turns a weak conclusion into strong assumptions.

**Question 20** RSN-08 · Chains of reasoning · spec Prf5

$a$ ,  $b$  and  $c$  are real numbers with  $a < b < c$ . Which of the following statements must be true?

**I**  $a + b < 2c$

**II**  $ab < bc$

**III**  $\frac{a+c}{2} < c$

**Answer: F**

I:  $a + b < c + c$ . III:  $\frac{a+c}{2} < \frac{c+c}{2} = c$ . II fails when  $b$  is negative:  $a = -3$ ,  $b = -2$ ,  $c = -1$  gives  $ab = 6 > bc = 2$ .

**Full method - how and why**

Inequality manipulation with 'must be true'. Two follow from replacing a term by a larger one; the third involves multiplying, which needs sign information.

**1** I: from  $a < c$  and  $b < c$ , add:  $a + b < 2c$ . **True.**

*why:* Adding two inequalities of the same direction is always valid.

**2** III:  $a < c$  so  $a + c < 2c$ , and dividing by 2 gives  $\frac{a+c}{2} < c$ . **True.**

*why:* Dividing by a positive constant preserves the direction. (The midpoint of two numbers lies below the larger.)

**3** II:  $ab < bc$  would follow from  $a < c$  only if we could multiply by  $b$  - valid only for  $b > 0$ .

*why:* Multiplying an inequality by a negative number reverses it.

**4** Counterexample:  $a = -3$ ,  $b = -2$ ,  $c = -1$  satisfies  $a < b < c$ , but  $ab = 6$  and  $bc = 2$ , so  $ab > bc$ . **False.**

*why:* All-negative values are the natural place to test a multiplication claim.

**5** So I and III.

*why:* Match to the list.

**Where it goes wrong**

- Accepting II by 'multiplying  $a < c$  by  $b$ ' without checking the sign of  $b$ .
- Testing only positive numbers, where II does hold.

**Takeaway** Adding inequalities is always safe; multiplying by a variable is not - you need its sign.